

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2018

HUM 1217
(Principles of Economics)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Explain the determinants of supply _____. (15)
(b) Solve the problems: (20)
There are 10,000 identical individuals in the market for commodity X, each with a demand function is given by $Q_{dx} = 12 - 2 P_x$ and 1,000 identical producers of commodity X, each with a supply function is given by $Q_{sx} = 20 P_x$.
(i) Obtain equilibrium price and quantity.
Now if the government decides to collect a sales tax of \$2 per unit sold from each of the 1,000 sellers,
(ii) What effect does this have on the equilibrium price and quantity?
(iii) Who actually pays the tax?
(iv) What is the total amount of tax collected by the government?
2. (a) Define price-elasticity of demand. Explain the various types of price-elasticity of demand with example. (15)
(b) The "Rahim Brothers" is a publisher of romance novels. The corporation hired an economist to determine the demand for its product. After months of hard work, the economist informed the company that the demand for the firm's novel is given by the following equation: (20)
 $Q_x = 12,000 - 5,000 P_x + 5 I + 500 P_c$
where,
 Q_x is the demand from "Rahim Brothers" novels, P_x is the price charged for the novels, I is the per capita income, P_c is the price of books from competing publishers. Assume that the initial values of P_x , I and P_c are \$5, \$10,000 and \$6 respectively. Using the above information, the company's manager wants to determine –
(i) What effect a price increase would have on total revenue.
(ii) How sales of the novels would change during a period of rising income.
(iii) The probable impact if competing publishers would raise their price.
3. (a) Explain Total Cost = Fixed Cost + Variable Cost (05)
(b) Define short run. For a given price explain how the competitive firm choose the level of output that maximize profit. (20)
(c) Explain the shut-down position of a firm under perfect competition. (10)
4. (a) What is production function? Explain the idea of factors of production. (15)
(b) Explain the idea of "law of diminishing returns" (10)
(c) Is land capital? Explain the idea of "Marginal Production" with an appropriate example. (10)

Section – B

5. (a) Explain the idea of Real GDP and Nominal GDP. Why do economists use real GDP rather than Nominal GDP to gauge economic well-being? (20)
(b) Explain the problems in calculating national income. (15)
6. (a) What is stock? What is bond? How are they similar? How are they different? (15)
(b) What is government budget deficit? How does it affect interest rates, investment and economic growth? (20)
7. (a) What is inflation? Explain the various types of inflation. (10)
(b) What are the main causes of inflation? (10)
(c) How does the government try to control inflation through monetary measure? (15)
8. (a) What do you mean by the word "productivity"? List and describe four determinants of workers' productivity. (15)
(b) A Speciality steel manufacturer produces two types of steel g_1 and g_2 . Type 1 requires 2 hours of melting, 4 hours of rolling and 10 hours of cutting. Type 2 requires 5 hours of melting, 1 hours of rolling and 5 hours of cutting. 40 hours are available for melting, 20 for rolling and 60 for cutting. The profit margin for type 1 is 24; for type 2 it is 8. Reduce the data to the equations and inequalities necessary to determine the output mix that will maximize profits. (20)
-

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2018

Math 1217
(Mathematics – II)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if missing any.

Section – A

1. (a) A function $f(x)$ is defined as follows: (13)

$$\begin{aligned} f(x) &= x && \text{for } 0 < x < 1 \\ &= 2x - 1 && \text{for } 1 \leq x \leq 2 \\ &= x - \frac{x^2}{2} && \text{for } x > 2 \end{aligned}$$

Discuss the continuity and differentiability of $f(x)$ at $x = 1$.

- (b) Differentiate $\tan^{-1}\left(\frac{\sqrt{1-x^2}-1}{x}\right)$ with respect to $\tan^{-1}x$. (09)
- (c) At a certain instant the edge of a cube is 5 cm and the volume is increasing at a rate of $2 \text{ (cm)}^3 / \text{minute}$. At what rate is the area of surface increasing? (13)
2. (a) If $y = a \cos(\log x) + b \sin(\log x)$, then find y_{n+2} by using Leibnitz's theorem. (10)
- (b) If $u = \log r$ and $r^2 = x^2 + y^2 + z^2$, then find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ in terms of r . (12)
- (c) Discuss the maxima and minima of $12(\log x + 1) + x^2 - 10x + 3$ (13)
3. (a) Define order and degree of a differential equation with example. (11)
Find the general solution of the differential equation
 $(6x - 2y - 7) dx - (3x - y + 4) dy = 0$.
- (b) Identify and solve the following differential equation : (11)
 $4y'' - 4y' - 3y = 0, y(0) = 1, y'(0) = 5$
- (c) Solve $(3x^2 y^4 + 2xy) dx + 2x^3 y^3 - x^2 dy = 0$ (13)
4. (a) Solve $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 3y = e^{-3x} + 2$ (11)
- (b) Determine the general solution of $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = x e^{2x} \sin x$ (11)
- (c) Solve $\frac{d^2 y}{dx^2} + y = \operatorname{cosec} x$ by the method of variation of parameters. (13)

Section – B

5. Answer any three of the followings : (35)

a) $\int \frac{(3x+2)}{(5x^2+2x+3)} dx$

b) $\int \frac{1-\sin x+\cos x}{1+\sin x-\cos x} dx$

c) $\frac{dx}{(1+x)\sqrt{1+2x+x^2}}$

d) $\int e^x \left(\frac{x^2+1}{(x+1)^2} \right) dx$

6. Determine three of the followings : (35)

a) $\int_0^\pi \frac{dx}{3+2\sin x+\cos x} dx$

b) $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx$

c) $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx$

d) $\int_0^1 \frac{x^3 \sin^{-1} x}{\sqrt{1-x^2}} dx$

7. (a) Obtain a reduction formula for $\int \sin^m x \cos^n x dx$. (12)

- (b) If $u_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$, then prove that $u_n + n(n-1)u_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$ and hence find (13)

the value of $\int_0^{\frac{\pi}{2}} x^5 \sin x dx$.

- (c) Calculate $\frac{dx}{(x^3-1)}$ (10)

8. (a) Represent the following series as a definite integral and hence evaluate : (10)

$$\lim_{n \rightarrow \infty} \left[\frac{1^2}{n^3+1^3} + \frac{1^3}{n^3+2^3} + \dots + \frac{n^2}{2n^3} \right]$$

- (b) Find the area of a region bounded by the parabola $y^2 = 4x$ and the straight line (13)

$$y = 2x - 4.$$

- (c) Evaluate $\int_0^\pi \frac{x \tan x}{\sec x + \cos x} dx$ (12)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Examination, 2018

URP 1211
(Urban Planning Principles)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures are not to scale. Do not draw given figures.

Section – A

1. (a) What do you mean by urban planning? Discuss the objective and importance of urban planning. (10)
(b) Draw your concept on urban area. How urban areas are defined in various countries based on minimum population criteria? (10)
(c) Describe various functional parts of an urban area with necessary example. (15)
2. (a) What do you mean by urban system? Describe the Rank Size rule and Primate City theory of urban system with necessary example. (10)
(b) What do you mean by the central place? Discuss the central place theory with necessary example. (10)
(c) Illustrate the procedures followed to select a site for establishing a commercial area. (15)
3. (a) Describe the concept- Eco-industrial park and Industrial ecology. (10)
(b) Describe various types of industries with necessary example. (15)
(c) Describe various parameters and elements of design that are considered while establishing an industrial park. (10)
4. (a) Based on function how roads are classified? Discuss with necessary example. (10)
(b) Discuss the relationships among urbanization and industrialization. (10)
(c) Discuss various types of zoning and their importance in urban planning and management with necessary example. (15)

Section – B

5. (a) "Urban planning is very crucial for a densely populated country like Bangladesh." Justify the statement and explain the characteristics of successful urban planning. (10)
(b) Define town centre. Describe the major function of a town centre.. (10)
(c) Write down the principles of creating a successful town centre and explain any two of them in brief. (15)
6. (a) Make a list of factors that must be considered for a sustainable residential development. (10)
(b) Define F.A.R. A new developed residential area of 42000 sq. m. has 35 plots of equal size. Owners of each plot are required to keep at least 150 sq. m. free of any sort of construction. Calculate the FAR applicable for that residential area (15)
(c) Draw a layout of a typical residential plot showing the various existing features on it. (10)
7. (a) Draw a schematic diagram showing different types of New Town. (10)
(b) Write down the major differences between Satellite Town and New Towns. (10)
(c) Mention three ways of tackling urban congestion. (07)
(d) Write short notes on the following types of recreational areas: (08)
i) Parkways or Boulevards ii) Green Belts
8. (a) Suppose you are working as a Town Planner at Khulna Development Authority (KDA). As a planner what factors should you consider for developing a new bus terminal in Khulna city? (15)
(b) Describe the facilities that must be present at a railway station used by commuters. (10)
(c) Draw a representative diagram showing minimum distance that any facility should be located from a station entrance measured along the pedestrian path. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Examination, 2018
URP 1251
(Introduction to Architectural History)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is Architecture? Is Architecture an Art? Justify your opinion. (10)
(b) What is Architectural Heritage? What are the influences on Architecture and Heritage? (10)
(c) Explain various influences on Egyptian architecture with necessary sketches and examples. (15)
2. (a) What is contemporary Architecture? (10)
(b) What are the architectural characteristics of high-tech Architecture? (15)
(c) What is "The New York Five"? (10)
3. (a) Draw the Master Plan of Greek Acropolis of Athens. (10)
(b) Explain the Architectural features of the Parthenon with its plan and section. (10)
(c) Explain the characteristics of Gothic Architecture with relevant sketches. (15)
4. (a) Write short notes on the following topics with necessary sketches (any 7). (35)
i) Greed Order
ii) Greek Optical Correction
iii) Greek Stoa
iv) Roman Road Construction
v) Aqueducts of Rome
vi) Flying Buttresses
vii) Renaissance Architecture Style
viii) Cubism
ix) Organic Architecture

Section – B

5. (a) "A street is a spatial entity and not the residue between buildings" – justify this as a planner. (10)
(b) Narrate the harmonious relationship between urban design and urban planning. (10)
(c) Do you think 'concentric ring theory' can be applicable to redesign a city like Khulna? If no, why and if yes, how. (15)
6. (a) Show the plan view of traditional Bengal ZamindarBari and point out the importance features in diagram. (10)
(b) 'Bungalow style evolved from the Bengal Architecture'- justify this statement. (13)
(c) Write down your understanding about the philosophy of Buddhist temple Architecture. (12)
7. (a) "Each Dome of a mosque provides a spiritual meaning". Describe this spiritual meaning. (10)
(b) Show the classification of temple according to their roofing system. (10)
(c) What are the major architectural features of Mughal Architecture period? Prove necessary diagrams. (15)
8. (a) Write short notes on the following (ay five - 5*7): (35)
i. Principles of sustainable architecture
ii. Patterns and calligraphy in Indo-Muslim architecture
iii. Traditional Rekha and Pidha style
iv. Curved ridge and gable roof
v. Buddhist Stupa
vi. Nirvana concept
vii. Buffer zone and its functions

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 2nd Term Regular Examination, 2018
URP 1281
(Surveying and Cartography)

Full Marks: 210

Time: 3.00 Hrs.

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

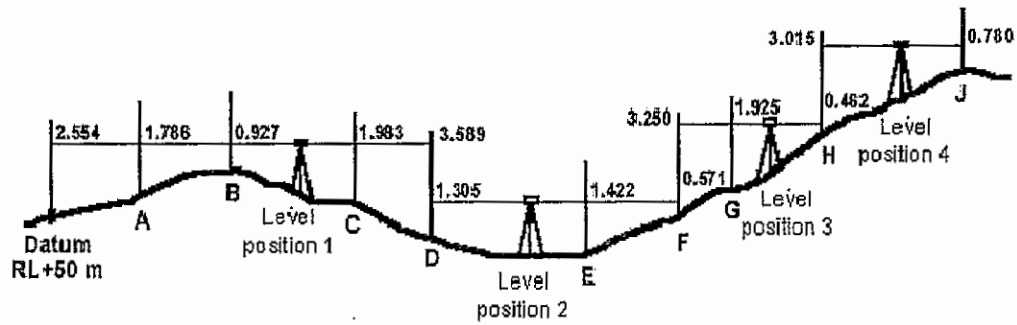
Section – A

1. (a) "Without maps we are spatially blind."- State your argument in favor or against this statement. (10)
(b) According to Rule of Thumb regarding map size, "If left to determine map size, produce it at the smallest size feasible". Explain with proper example to justify the statement. (10)
(c) What do you understand by thematic maps? Classify thematic maps and provide a brief description of each with necessary figures. (15)
2. (a) "Distortions in maps make geographers S.A.D.D."- Explain the statement with relevant examples of various types of distortions. (10)
(b) Write about Universal Transverse Mercator (UTM) with respect to Bangladesh. Why is BTM projection system used in Bangladesh instead of having global projection system (UTM)? (15)
(c) i) Two sites 2.5 inches apart on a map are actually 750 miles apart. Find out the scale at which the map has been drawn. (10)
ii) Two bus-stops 800m apart measures 2cm on a map. The map scale is 1: n. Find the value of n.
3. (a) What are the different types of mapping elements? Explain the relationship between surveying and cartography with relevant example. (10)
(b) Define R.F. An area of 8 sq. inches in drawing represents an area of 512 sq.km. What is the R.F of the scale? (15)
(c) What do you mean by large scale and small scale maps? Two maps are drawn at scales of 1:24573 and 1:38946 respectively. Which of the two maps will show a greater degree of detail and why? (10)
4. (a) Annotate any three of the following with proper examples (if it requires) (15)
(i) Geoid (ii) Ellipsoid or Spheroid (iii) Locator Map (iv) Topographic Map
(v) Planimetric Map
(b) Show the supporting diagrams how maintaining good sight lines can help to prepare a good map layout. (10)
(c) Explain the role of Visual Balance in obtaining a good map layout. (10)

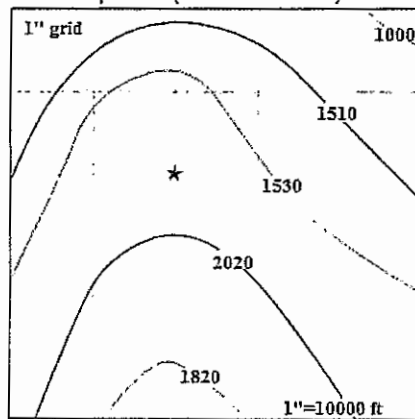
Section – B

5. (a) What is offset? Mention some guidelines for number of offsets taken in the field for different object nature. Use necessary sketch. (15)
(b) A close traverse is conducted with five stations A, B, C, D and F taken in anticlockwise order, in the form of a regular pentagon. If the FB of AB is 30°0', find the FBs of the other sides. (15)
(c) On an old map a line was drawn to a magnetic bearing of N39°30'W, when the declination was 3°30' W. Find the present bearing of the line, if the declination is 4°15' E. (05)
6. (a) What are the points to be kept in mind while selecting survey stations during chain survey? (05)
(b) What is orientation? Describe the methods of orientation with sketch. (15)
(c) Describe the procedure of setting up the plane table over a station. (15)

7. (a) The following figure is the outcome of leveling survey. Calculate the elevations using collimation and rise-and-falls method and check your result. (15)



- (b) Explain the characteristics of contour with necessary sketches. (10)
 (c) Define the following with necessary sketch and example: (10)
 i) Whole circle bearing and reduced bearing
 ii) Fore bearing and back bearing
 iii) Magnetic declination
8. (a) Determine the elevation of the point (star marked) from the following figure. (10)



- (b) Mention some applications of GPS with examples. (15)
 (c) What are the advantages and disadvantages of total station? (10)